📑 Basic Information:

Experiment No: 08

Date:

Title: Design and Implementation of a Java Employee Salary Management Program Using Interfaces to Calculate and Display Salary Details for Regular and Contract Employees.

🎯 Objective /Aim:

To write a Java program that uses an interface to calculate and display salary details for regular and contract employees based on their employee ID.

🛠️ System Requirements:

Hardware:

-> Standard PC/Laptop

-> Intel Core i3/i5 processor or equivalent

-> Minimum 4 GB RAM

-> Minimum 500 MB free

Software:

->Operating System: Windows / Linux / macOS

-> Java Development Kit (JDK 17 or JDK 21)

-> Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code

-> Command Prompt / Terminal

Algorithm/Logic Description:

1. Start the program.
2. Define an Employee interface with a salaryDetails() method.
3. Create RegularEmployee and ContractEmployee classes that implement the interface.
4. In each class, calculate the total salary and display the salary components.
5. In main, read the employee ID.
6. If the ID starts with R, create a RegularEmployee object and display its salary details.

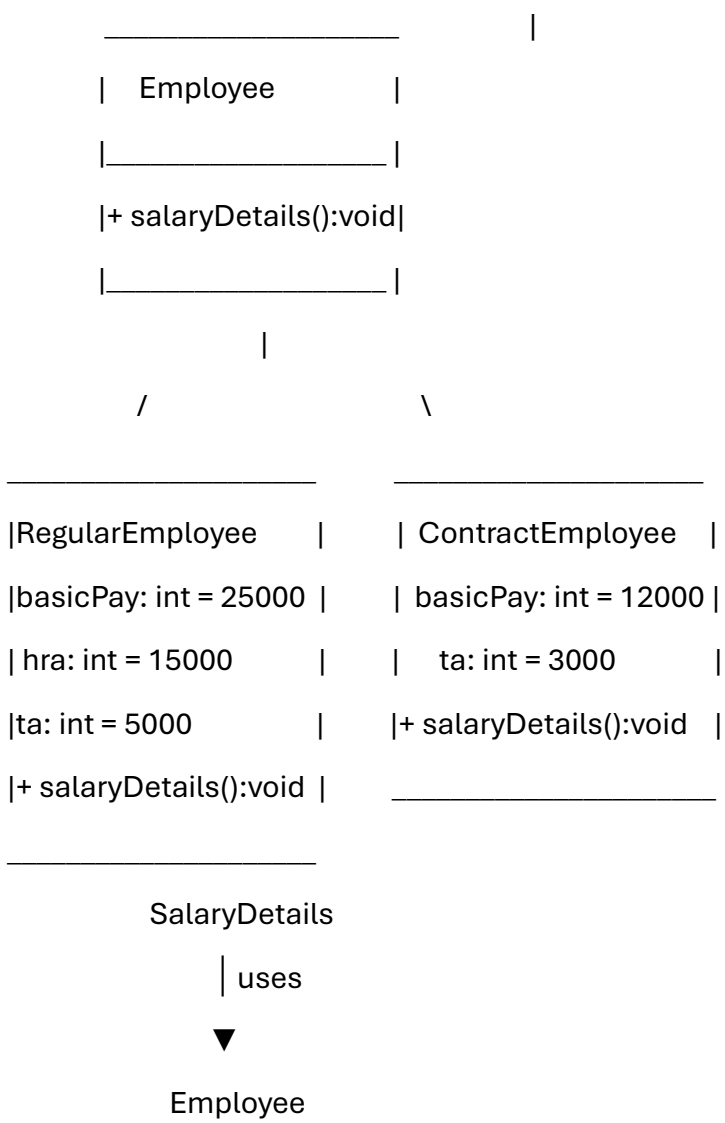
7. If the ID starts with C, create a ContractEmployee object and display its salary details.

8. Otherwise, display “Invalid Employee Id.”

9. Close the scanner.

10. Stop the program.

UML Class Diagram:



Source Code:

```
package ride;

import java.util.Scanner;

// Interface that defines the salary details method

interface Employee {
```

```
    void salaryDetails();
}
// Regular employee with basic pay, HRA, and travel allowance
class RegularEmployee implements Employee {
    int basicPay = 25000;
    int hra = 15000;
    int ta = 5000;
    // Calculates and displays the regular employee's salary
    @Override
    public void salaryDetails() {
        int total = basicPay + hra + ta;
        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("HRA: " + hra);
        System.out.println("T.A: " + ta);
        System.out.println("Total Amount: " + total);
    }
}
// Contract employee with basic pay and travel allowance
class ContractEmployee implements Employee {
    int basicPay = 12000;
    int ta = 3000;
    // Calculates and displays the contract employee's salary
    @Override
    public void salaryDetails() {
        int total = basicPay + ta;
        System.out.println("Salary Details:");
        System.out.println("Basic Pay: " + basicPay);
        System.out.println("T.A: " + ta);
```

```
        System.out.println("HRA: No HRA");
        System.out.println("Total Amount: " + total);
    }
}
// Main class that selects the employee type from the ID
public class SalaryDetails {
    public static void main(String[] args) {
        // Scanner reads the employee ID from the keyboard
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Employee Id: ");
        String id = sc.nextLine();
        // Interface reference can refer to either employee type
        Employee employee;
        // IDs beginning with R are regular employees
        if (id.startsWith("R")) {
            employee = new RegularEmployee();
            employee.salaryDetails();
        }
        // IDs beginning with C are contract employees
        else if (id.startsWith("C")) {
            employee = new ContractEmployee();
            employee.salaryDetails();
        }
        // Any other ID is invalid
        else {
            System.out.println("Invalid Employee Id");
        }
        // Close the scanner after input is complete
        sc.close();
```

```
    }
}
```

Relationship:

RegularEmployee implements the Employee interface.

ContractEmployee implements the Employee interface.

SalaryDetails creates and uses either employee class based on the ID.

Output Screenshot:



Result:

The program uses the Employee interface to display salary details for regular and contract employees. It identifies the employee type from the ID prefix and calculates the total salary: 45,000 for a regular employee and 15,000 for a contract employee.